Team Name - T123

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Github link - `https://github.com/Anoop-Parashar/COL7361-Assignments`

Assignment 2
COL761 2026
Q1 k-means clustering in high-dimensions

Experimental Setup and Implementation

We implemented K-Means in Python using NumPy and scikit-learn. For each dataset, inertia was computed for $k \in \{1, \ldots, 15\}$ and the elbow point was automatically detected using Yellowbrick's `KElbowVisualizer`; the resulting plots were saved as `plot.png`.

(a) Selected Value of k

The elbow plot (Figure 1) shows the K-Means objective function — the total within-cluster sum of squared distances — as a function of k for k ∈ {1, 2, ..., 15}. The Kneedle algorithm (implemented via Yellowbrick's KElbowVisualizer) was used to identify the elbow point programmatically as the point of maximum curvature on the objective curve.

We got an optimal value of k as 3 for dataset1, and for dataset2, also 3. The plots that we got are the following:

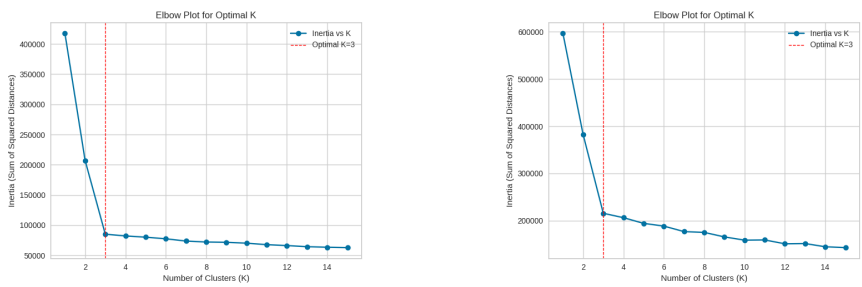

Figure 1: Figure a.1 Plot for dataset1 and Figure a.2 Plot for dataset2

The optimal value of k helps us determine the optimal number of clusters, which differentiates the points into fine-grained clusters with the right value of the within-cluster sum of squares. The k is usually the turning point in the plot. The kneedle algorithm eliminates the need to manually search for the optimal point on the plot. It uses the curvature detection of the plot. It finds the optimal point by locating the point furthest from the line connecting the start and end points of the plot. By finding the optimal number of clusters, we balance between too many and too few clusters. It uses Spline interpolation, which removes the noise in the plot. This prevents printing false optimal points.

The kneedle algorithm defines a straight line L connecting the first and last points. It calculates the perpendicular distance between each point on the curve and Line L. The point with the largest distance is reported as the optimal point. The optimal point is where a steep vertical drop meets a flat horizontal plateau.

(b) Structural and Statistical Assumptions of K-Means

Euclidean Space: The k-means algorithm assumes that the similarity between the objects is measured using the Euclidean distance function.

- The k-means assumes the clusters are centered, spherical, and isotropic.
- The k-means assumes that the data is represented in a space where Euclidean distance remains a meaningful measure of similarity. In very high-dimensional spaces, this assumption is often violated as distances between all points tend to converge (distance concentration)
- It assumes that cluster sizes are homogeneous, i.e., they are of equal volume and density.
- It assumes the clusters are not non-convex. It does not work on nested circles.

Statistical assumption: The user knows the correct value of k, the number of clusters in the dataset, beforehand and provides it as input. It assumes that each object belongs to only one cluster. It uses the greedy iterative approach. It randomly selects k objects as the starting points and uses them to find the clusters.

(c) Limitations of the Model Selection Procedure

- If the curve does not have a bend, the algorithm will provide a point that does not correspond to a meaningful cluster.
- The optimal value of k produced depends on the range of values tested.
- The difference curve might have multiple peaks. The Kneedle might pick the first one it finds, missing the global peak.
- If the sensitivity parameter is too high, the algorithm may not find the peak and might give noise as the output.
- The method only works for strictly decreasing and convex objective functions.
- In high-dimensional datasets, the drop in inertia is not present. This leads to no elbow in the plot. Hence, the algorithm fails.
- The furthest point can be biased if the initial drop from k=1 to k=2 is very large.

(d) Methodological Improvements and Alternative Validation

- Improvement in the initialization of the random centroids. We can use K-means++ to improve the selection of initial centers.
- We can use silhouette analysis, which assesses both cohesion and separation. This method provides more granular validation by evaluating the quality of each point's assignment.
- We can use Gaussian mixture models, which use ellipsoidal, isotropic clusters and provide soft clustering, rather than considering only spherical, convex clusters in the K-nearest-neighbor model.
- We can use K-medoids, which uses actual data points rather than averages, and is more robust to noise and extreme values.

(e) When K-Means is Not Appropriate: Failure Modes

Failures: K-Means encounters three main issues: (1) Non-convex geometry — decision boundaries are Voronoi regions (linear), so interleaved or ring-shaped clusters like two moons or concentric circles cannot be separated; (2) Density imbalance — since the objective minimizes total inertia, large sparse clusters are split and small dense clusters merge, as the centroid gravitates toward high-density areas; (3) Curse of dimensionality — in high-dimensional space, pairwise distances tend to converge ($\frac{d_{max}-d_{min}}{d_{min}} \to 0$), rendering the Euclidean centroid meaningless and causing the elbow method to disappear.

Principled Detection Procedure:

- **Cluster Tendency:**We will calculate the Hopkins Statistic (H). If $H \approx 0.5$, the dataset is uniformly distributed; a value near to 1.0 confirms the presence of meaningful, non-random clusters.
- **Inertia Ratio:** We will compare the ratio of within-cluster inertia to total inertia (W/T). A high ratio suggests that K-means is failing to capture significant variance, which signals that the spherical assumption is invalid.
- Project your data using PCA or T-SNE into 2D or 3D. If we see moons, long streaks, or rings, then k-means is likely the wrong choice.
- Use the Silhouette plot; if cluster sizes are uneven or many points have a negative silhouette width, then it suggests that the spherical assumption is forcing a bad fit.
- Use k-means with different random seeds at each iteration. If there is a drastic shift in the resultant boundaries in the run then the data likely lacks the distinct, compact centres that K-means requires.

References

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